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A reduced Composite Indicator for Digital Divide measurement at the regional level: An application to the Digital Economy and Society Index (DESI)

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ABSTRACT

The present paper investigates the problem of defining an easy-to-implement tool to evaluate the Digital Divide (DD) among and, above all, within countries to grasp sub-national divides and appropriately inform policy-makers guiding digital transformation processes. With this aim in mind, the study focuses on the Digital Economy and Society Index (DESI), the most recent and currently used CI for DD measurement across European states. A methodology employing correlation and principal component analysis is adopted to assess redundancies in the DESI and obtain a reduced version of the index with a significantly lower number of indicators. Experiments performed using the 2020 data related to the 29 EU countries demonstrate that the reduced DESI produces very similar results to the original one, thus proving the existence of redundancies in its current structure. An extensive analysis of the Italian case shows how the reduced version of the index can be more easily applied at the sub-national level, thus helping to detect DD between regions and offering meaningful policy-making support. In particular, a framework to analyze the obtained results is proposed to support the identification of adequate interventions at the national level or targeted actions to mitigate digital disparities in lagging regions.

1. Introduction

The rapid development of Information and Communication Technologies (ICTs) can be considered one of the fundamental factors fueling growth and transforming the way business is conducted, driving the digital transformation of the economy and society (Cruz-Jesus et al., 2012, 2017). However, a striking and alarming disparity in ICTs access and usage among and within countries might hinder this transition and prevent it from achieving inclusive development. The term Digital Divide (DD) has been coined to describe disparities in access and use of ICT and the ability of individuals and economic systems to participate in the global information society (Liao et al., 2022).

Due to its importance, DD has gained increasing academic attention over the years. The recent review by Lythreatis et al. (2022) provides a comprehensive overview of the topic, also shedding light on emerging factors that are becoming crucial for describing DD (e.g., data inequalities, type of internet access, algorithmic awareness). In general, the authors note that existing works investigated DD from two different perspectives: (i) the detection of factors and dimensions influencing the phenomenon (Bucea et al., 2020; Pontones-Rosa et al., 2021); (ii) the study of the multidimensionality of DD and of its measurement (Hidalgo et al., 2020; Arakpogun et al., 2020; Shakina et al., 2021, Mubarak, 2015). The latter is the focus of the present work.

From a methodological point of view, DD is typically measured through Composite Indicators (CIs), defined according to a hierarchical framework to account for its intrinsic multi-criteria structure (i.e., articulated in categories and, often, sub-categories). Even if CIs are widely exploited as useful tools in policy analysis, their appropriate methodological design is fundamental to avoid potential misrepresentation or misinterpretation (OECD, 2008). As we show in the next section, several institutions and organizations have been developing proposals for DD measurement, usually based on complex conceptual models requiring the aggregation of many indicators. Such an approach seems to respond to the assumption that a higher number of indicators should result in more accurate CIs. However, as the literature highlights, this can often lead to the consideration of multiple highly correlated indicators (redundancy), which may generate a distortion of the final results (ranking or rating) with the consequent misrepresentation/

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misinterpretation of the phenomenon (see Bruno et al., 2011; James, 2012; Gerpott and Ahmadi, 2015a, 2015b; Park et al., 2015; Milenkovic et al., 2016). Besides, a high number of indicators may also undermine the data collection process, making CIs' applicability in other geographical contexts (*inter-country*) and levels (*intra-country*) rather critical. In turn, this may hinder: (i) a homogenous evaluation of DD at the national and local levels; (ii) the identification of local geographical patterns of inequalities in DD (as underlined by Lythreatis et al., 2022); (iii) the deployment of targeted and informed policies at the local level.

Therefore, the availability of a lean and reliable methodology for DD measurement is a major policy concern, whose importance has been reinforced by the recent COVID-19 pandemic. In fact, by accelerating the digital transformation of services, it has dramatically highlighted how DD may affect and exacerbate access disparities to essential services (Lai and Widmar, 2021), such as healthcare assistance (Ramsetti and Adams, 2020), education (Iivari et al., 2020; Soomro et al., 2020) or e-government services (Pérez-Morote et al., 2020). In this context, the need for performance evaluation and benchmarking emerged at the country level to assess the magnitude and the evolution of DD, keep track of the effectiveness of proposed interventions and provide rankings that could be easily divulgated on media. Notably, digitalization is also one of the pillars of the Next Generation EU, the economic recovery plan introduced by the European Commission in July 2020 to support countries affected by the COVID-19 pandemic. For this reason, tools for measuring DD and monitoring progress among and within countries are of timely relevance.

In such a context, this paper analyzes the *Digital Economy and Society Index* (*DESI*), the most recent CI proposed by the European Community to evaluate and monitor the DD among European countries. As we detail next, it has a relatively complex structure due to the number and the variety of included indicators. Therefore, the main goal of this work is to verify whether DESI suffers from redundancy and, accordingly, to define a "reduced" and more easily implementable version of the index using a relatively simple methodological approach. The devised methodology employs correlation and Principal Component Analysis (PCA) to drive the selection of a restricted number of indicators to be included in the DESI calculation, assuring an effective replication of the results produced by the actual version of the index. Besides, through an application to the Italian regions, we also demonstrate how the slimmer and agile structure of the CIs we propose ease its application at different geographical levels, overcoming missing data issues and offering meaningful insights and policy-making support.

The present study aims at giving a twofold contribution. Firstly, it can contribute to the literature by showing how a relatively simple methodological approach can lead to slimmer - but empirically robust - versions of traditional indicators for DD measurement. Note that, due to its generality, the proposed approach can be applied to other indexes and their future releases. In this regard, it is worth highlighting that, to the best of the authors' knowledge, there is no work dealing with the specific CI at hand (i.e., the DESI). Secondly, and more ambitiously, this study also aims at showing how easier-to-implement indexes can contribute to the practice of policy-making, as they could likely favor a homogenous evaluation and monitoring of national and local policies against guidelines, benchmarks and targets set at the European level.

The remainder of the paper is organized as follows. In Section 2, we present the main CIs proposed for DD measurements, with a particular focus on the DESI. In Section 3, we describe the adopted methodology for the statistical analysis and the definition of the reduced version of the index, whose results are discussed in Section 4. In Section 5, we show the application of the obtained reduced index to the Italian regions and we discuss the policy implications of the analysis. Finally, Section 6 closes the paper with a summary of the work done, its limitations, and directions for future research.

2. Composite Indicators for Digital Divide measurement

As noted in the previous section, various CIs have been proposed for DD measurement by policy-making institutions and think-tank organizations. Here, we first briefly present the main proposals known to date (Section 2.1). Afterwards, a more detailed description of the DESI follows in Section 2.2.

2.1. General overview

This section aims to provide the reader with a general overview on DD measurement - and its evolution over time - through a brief presentation of the main CIs proposed for this purpose. We summarize them in Table 1, which reports, for each CI, the release year, the categories (in bold) and sub-categories (in brackets, if any) and the reference paper describing the related methodological note. For the sake of brevity, we do not inform on the specific indicators by category or sub-category, but, to this end, interested readers can refer to the references indicated in the table.

The first CIs were introduced in 2001 (*Technology Achievement Index* – TAI, and *Networked Readiness Index* - NRI) when DD was a relative new phenomenon, while the last one was released in 2014 (DESI). Despite the first glance differences among CIs, a closer look at their structures reveals that these are more formal than substantial, being often limited to nomenclature issues. Indeed, almost all the indexes share, to more or less large extents, three main aspects describing the phenomenon: (i) *ICT infrastructure development*, focusing on connection networks infrastructures, the quality of internet connection, and users' accessibility to the service; (ii) *ICT skills*, dealing with users' skills and knowledge of handling computer devices and Internet services; (iii) *ICT usage*, emphasizing the different uses of internet services made by single users, private companies and public organization.

The Table clearly shows that the DESI is the most complete and complex CI, given the high number and variety of included indicators. Also, as already highlighted, it is currently used by the European Commission for DD measurement and policy-aiding purposes across the Member States. Its main structural and methodological features are described next.

2.2. The DESI

The Digital Economy and Society Index (DESI) measures EU countries' progress towards a digital economy and society. Since its introduction in 2014, its framework has been updated yearly. Here, we refer to its 2020 release. The DESI combines a relevant number of indicators (37 in total) classified into five categories and 12 sub-categories, which we report in Table 2. For the sake of clarity, (sub-)categories and indicators are labeled with a unique ID code.

Compared with previous proposals, the DESI not only extends "more classical" DD domains (*Connectivity, Human Capital, Use of Internet Services*) by considering up-to-date technologies and more refined elements, but also includes novel up-to-date categories, namely *Integration of Digital Technology* and *Digital Public Services*. For example, as concerns traditional infrastructural aspects, the DESI category *Connectivity* considers the evolution of ICT technology, by including indicators related to mobile and fixed *broadband coverage*, *take-up* and price *indexes*. *Human Capital* essentially focuses on users' skills, distinguishing between basic and advanced competencies (*Internet User Skills* and *Advanced Skills and Development*, respectively). Interestingly, the latter also emphasizes gender balancing issues (under the indicator *Female ICT specialists*). The *Use of Internet Services*, traditionally referred to the internet usage in practice (*Internet use*), is extended to account for different profiles of users, differentiated by the conducted types of *Activities* and *Transactions*. The category *Integration of Digital Technology* accounts for DD measurement from the perspective of digital transformation into *Businesses* and *e-Commerce*. Considered indicators assess, among the others,

Table 1
List and characteristics of the considered CIs for the DD measurement.

	Index	Release year	Number of categories (sub-categories)	Categories (sub-categories)	Reference
TAI	Technology Achievement Index	2001	4 (0)	**Creation of technology, Diffusion of recent innovations, Diffusion of old technologies, Human skills development**	Desai et al. (2002)
NRI	Networked Readiness Index	2001	4 (12)	**Environment** (Political and regulatory, Business and Information), **Readiness** (Infrastructure, Affordability, Skills), **Usage** (Individual, Business, Government), **Impact** (Economic, Social)	Dutta and Lanvin (2019)
DAI	Digital Access Index	2003	5 (0)	**Infrastructure, Affordability, Knowledge, Quality, Usage**	ITU (2003)
Infostate	Infostate Index	2003	2 (4)	**Infodensity** (Networks, Skills), **Info-use** (Uptake, Intensity)	Orbicom (2005)
ICTOI	ICT- Opportunity Index	2005	2 (4)	**Infodensity** (Networks, Skills), **Info-use** (Uptake, Intensity)	ITU (2007a)
DOI	Digital Opportunity Index	2005	3 (0)	**Opportunity, Infrastructure, Utilization**	ITU (2007b)
ICTDI	ICT Diffusion Index	2006	2 (0)	**Connectivity, Access**	UNCTAD (2006)
IDI	ICT Development Index	2009	3 (0)	**Access, Use, Skills**	ITU (2016)
DESI	Digital Economy and Society Index	2014	5 (12)	**Connectivity** (Fixed Broadband take-up, Fixed broadband coverage, Mobile broadband, Broadband price index) **Human capital** (Internet User Skills, Advanced Skills and Development), **Use of Internet services** (Internet use, Activities online, Transactions), **Integration of Digital technology** (Business digitalization, e-Commerce), **Digital public services** (e-Government)	European Commission (2020)

the levels of *social media* and enabling Industry 4.0 technologies' usage in companies (*big data* and *cloud*), the overall *turnover* of the e-Commerce sector, and its degree of diffusion within *SMEs*. Finally, *Digital Public Services* evaluates digital penetration in public administrations through, for instance, the number of actual *digital services' users*, the availability of *Digital Services for Business* and *Open Data*.

According to the methodological note released by the European Commission (2020), the DESI is calculated as a weighted sum, using the weights associated with each category, sub-category, and indicator indicated in Table 2.[1] Moreover, the same reference specifies that, for the correct calculation of the index, each indicator must: (i) be collected regularly; (ii) be relevant for a policy area of interest; (iii) not be redundant. Following this third condition, the DESI should not contain overlapping indicators, neither statistically nor in terms of meaning, to avoid distortion of the produced scores and minimize applicability issues due to data collection. For this reason, we select this index as a suitable case study and set it at the core of the developments presented in the next sections.

3. Methodology

This section describes the methodology used for the statistical analysis of the DESI and the definition of its reduced version. Taking inspiration from various studies showing successful applications of data reduction and redundancy assessment strategies (e.g., Bruno et al., 2011; Genovese et al., 2017; Brint et al., 2021, to name a few), we structure our analysis according to the following four steps:

1. Correlation analysis between pairs of indicators;
2. Application of Principal Component Analysis (PCA);
3. Indicators' selection and computation of the reduced index;
4. Verification process.

[1] A min-max normalization is applied. Chosen minimum and maximum values for normalization are fixed and available in the DESI methodological note. Thus, indicators range between 0 and 1, and their scores are expressed in percentages.

We detail each phase in the following sections. To this end, some useful notation is introduced. With reference to a generic CI, let I be the set of its indicators, grouped into a set C of categories and a set S of sub-categories. Accordingly, we denote by S_c the set of sub-categories belonging to category $c \in C$, and by I_s the set of indicators belonging to sub-category $s \in S$. Additionally, let κ_c, λ_s, γ_i be the weights associated with the generic category $c \in C$, sub-category $s \in S$ and indicator $i \in I$, respectively. Clearly, it must be verified that: (i) $\sum_{c\in C}\kappa_c = 1$; (ii) $\sum_{s\in S_c}\lambda_s = 1$, $c \in C$; (iii) $\sum_{i\in I_s}\gamma_i = 1$, $s \in S$.

In what follows, steps 1–3 are separately performed for each category. This choice is motivated by the fact that we want to keep the main framework of the DESI unchanged. Indeed, since the single categories have been conceived to capture different aspects of the phenomenon under analysis, we are not going to eliminate any of them. Instead, with reference to the single category (or domain), we want to detect if potential redundancies exist within the cluster of indicators used to measure it.

3.1. Correlation analysis between pair of indicators

For each category $c \in C$, we calculated the correlation matrix, say $P_c = \{\rho_{ij}\}$, whose generic element ρ_{ij} is the correlation between the generic pair of indicators (i,j) belonging to category $c \in C$ (with $(i,j) \in I_c, j > i$). Note that $\frac{|I_c|\times(|I_c|-1)}{2}$ values should be computed. This preliminary analysis is useful to assess the general level of redundancy in the initial data set.

3.2. Application of Principal Component Analysis (PCA)

PCA is a multivariate technique, commonly used for dimensionality reduction purposes, that, starting from a set of correlated variables, seeks to build a new set of uncorrelated artificial variables, called Principal Components (PCs – Kim and Mueller, 1978a, 1978b; Stevens, 1986). Here, PCA is applied to each category $c \in C$, taking as input the corresponding set of indicators I_c. We denote the resulting set of PCs associated to category c by B_c. Specifically, the generic PC b_k is obtained as a linear combination of the input variables and defined as: $b_k = \sum_{i\in I_c} a_{ki} i$. In the latter expression, the vector $a_{k1}, \ldots, a_{k|I_c|}$ is represented by the *eigenvectors* of the *covariance matrix* of the data, corresponding to

Table 2
The DESI framework (European Commission, 2020).

Category [ID – Name (Weight)]	Sub-category [ID – Name (Weight)]	Indicators [ID – Name (Weight)]
1 - Connectivity (0.25)	1a - Fixed Broadband take-up (0.25)	1a1 - Overall fixed BB take-up (0.500)
		1a2 - At least 100 Mbps fixed BB take-up (0.500)
	1b - Fixed broadband coverage (0.25)	1b1 - Fast BB (NGA) coverage (0.500)
		1b2 - Fixed VH Capacity Network coverage (0.500)
	1c - Mobile broadband (0.35)	1c1 - 4G coverage (0.250)
		1c2 - Mobile BB take-up (0.250)
		1c3 - 5G readiness (0.500)
	1d - Broadband price index (0.15)	1d1 - Broadband price index (1.000)
2 - Human capital (0.25)	2a - Internet User Skills (0.50)	2a1 - At least basic digital skills (0.333)
		2a2 - Above basic digital skills (0.333)
		2a3 - At least basic software skills (0.333)
	2b - Advanced Skills and Development (0.50)	2b1 - ICT specialists (0.333)
		2b2 - Female ICT specialists (0.333)
		2b3 - ICT graduates (0.333)
3 - Use of internet services (0.15)	3a - Internet use (0.25)	3a1 - People who never used internet (0.500)
		3a2 - Internet users (0.500)
	3b - Activities online (0.50)	3b1 - News (0.166)
		3b2 - Music, video and games (0.166)
		3b3 - Video on demand (0.166)
		3b4 - Video calls (0.166)
		3b5 - Social networks (0.166)
		3b6 - Doing an online course (0.166)
	3c – Transactions (0.25)	3c1 - Banking (0.333)
		3c2 - Shopping (0.333)
		3c3 - Selling online (0.333)
4 - Integration of digital technology (0.20)	4a - Business digitalization (0.60)	4a1 - Electronic information sharing (0.167)
		4a2 - Social media (0.167)
		4a3 - Big data (0.333)
		4a4 - Cloud (0.333)
	4b - e-Commerce (0.40)	4b1 - SMEs selling online (0.333)
		4b2 - e-commerce turnover (0.333)
		4b3 - Selling online cross-border (0.333)
5 - Digital public services (0.15)	5a - e-Government (1.00)	5a1 - e-government users (0.200)
		5a2 - Pre-filled forms (0.200)
		5a3 - Online service completion (0.200)
		5a4 - Digital public services for business (0.200)
		5a5 - Open data (0.200)

the $|I_c|$ *eigenvalues*.

In order to select a significant subset $B_c^{'}$ of PCs ($B_c' \subset B_c$), various rules can be used. In our study, we adopt the cumulative percentage threshold of the total explained variance (see Jolliffe, 2002). In practice, PCs are ranked in non-increasing order of their explained variances within the data set. Then, the first PCs are selected such that the cumulative percentage variance is higher than or equal to a given threshold ψ, which is a user-defined parameter. We underline that the number p of selected PCs, i.e., the number of indicators included in the revised index, is strongly dependent on the parameter ψ ($p = p(\psi)$) and, hence, its effective calibration is a crucial aspect of the analysis.

3.3. Indicators' selection and computation of the reduced index

As PCs are linear combinations of the original indicators, they have no physical meaning. Hence, we calculate the correlation matrix $R_c = \{r_{ib}\}$ between each indicator $i \in I_c$ and each selected PC $b \in B_c^{'}$. Then, we select for each PC b, the indicator $i_b^{\star} \in I_c$ showing the highest correlation with b itself $\left(i_b^{\star} = \underset{i \in I_c}{\operatorname{argmax}}\{r_{ib}\}, b \in B_c^{'}\right)$. Of course, each indicator is selected without repetition. This way, we identify the subset I_c' ($I_c' \subset I_c$) of unique indicators, explaining the given percentage ψ of the cumulative variance. Again, this step is repeated for each category $c \in C$.

At this point, all the selected indicators (i.e., $\cup_{c \in C} I_c'$) are aggregated, according to the original rule (e.g., weighted sum method), to obtain the reduced version of the index. The weights associated to the deleted indicators are redistributed across the undeleted ones within the same subcategory, proportionally to their original weights. For instance, let us denote with I_s' the subset of indicators selected in the reduced version of the index, belonging to the generic sub-category $s \in S_c$. The new weight γ_i^{new} associated to the generic indicator $i \in I_s'$ is given by the sum of its original weight (γ_i) plus the quota received from the redistribution of the weight associated to the deleted indicators in the same subcategory ($\sum_{t \in I_s \setminus I_s'} \gamma_t$), according to the following formula:

$$\gamma_i^{new} = \gamma_i + \sum_{t \in I_s \setminus I_s'} \gamma_t \times \frac{\gamma_i}{\sum_{k \in I_s'} \gamma_k} \quad i \in I_s'$$

If a given sub-category is completely deleted, the associated weight is redistributed across the other subcategories. Also note that, since original categories are kept in the new version of the index, their weights (κ_c) remain unchanged.

Clearly, an alternative approach would be to avoid considering subcategories, and solely focus on indicators. This would imply redistributing weights of deleted indicators across the selected ones belonging to the same category. Although we do not report on this case, we emphasize that both the approaches led to almost identical results as those shown in Section 4.

3.4. Verification process

To verify the capability of the revised CIs to reproduce the original results, in terms of scores and ranking, we computed the *correlation index* and the *Spearman's Footrule* (Diaconis and Graham, 1977). In the first case, we calculate the correlation coefficient between the scores produced by the reduced indexes and the original ones. In the second case,

we compare the final rankings they provide. Let us denote by M the set of elements of the data set and by R, R' the rankings produced by the original and the revised CIs, respectively. Also, let $\sigma_R(m)$, $\sigma_{R'}(m)$ the positions of element $m \in M$ in the two rankings. The Spearman's Footrule (*SF*) can be calculated as:

$$SF(\sigma_R, \sigma_{R'}) = \sum_{m \in M} |\sigma_R(m) - \sigma_{R'}(m)|$$

which varies from 0 (the two rankings are identical) to $\frac{1}{2} Z^2$ (the maximum possible value). Then, the normalized version of the Spearman's Footrule (*NSF*) is obtained as:

$$NSF(\sigma_R, \sigma_{R'}) = 1 - \frac{2 \sum_{m \in M} |\sigma_R(m) - \sigma_{R'}(m)|}{Z^2}$$

which continuously varies between 0 (rankings are completely different) and 1 (rankings are identical).

4. Empirical analysis

In this section, we show the results obtained through the application of the described methodology. To this end, we used the 2020 data available for the 29 EU countries (including the UK), retrieved from official online sources (https://digital-agenda-data.eu/datasets/desi). Results obtained for each methodological step are presented in the next sections. The methodology was implemented using R version 4.1.1 and RStudio version 2021.09.0 (tidyverse package).

4.1. Step 1 - Correlation analysis

Recall that, in this step, we calculate correlations for each pair of indicators belonging to the same category. Results are summarized in Table 3, which reports, for each of the five categories of the DESI, the corresponding number of pairs showing a correlation value higher than or equal to a given threshold $\overline{\rho}$ ($\overline{\rho} \in \{0.60, 0.70, 0.80, 0.90\}$).

As the table shows, the number of highly correlated pairs of indicators is not negligible. Indeed, about 30 % of them (41 out of 129) have a correlation at least equal to 0.60. Although this number tends to decrease as $\overline{\rho}$ increase, one can still notice four pairs of very strongly correlated indicators (≥ 0.90) belonging to the *Human Capital* and *Use of Internet Services* categories. A more detailed look into the latter is provided by their correlation matrixes in Tables 4 and 5.

If we focus on the *Human Capital* category (Table 4), more than half of the possible pairs have a correlation value higher than 0.60. In particular, three pairs exceed 0.90. Specifically, the indicator *At least basic digital skills* practically coincides with the indicator *At least basic software skills* (correlation equals 1.00), and is strongly correlated with *Above basic digital skills* (0.94). In turn, the latter exhibits a very high correlation with *At least basic software skills* (0.92). These results suggest that people with "basic" digital skills also have "basic" software skills and "above" digital skills, indicating that only one of these indicators could potentially explain all the information related to non-specialist users' abilities in ICT usage.

Table 3
Pairs of correlated indicators within the same category.

Categories	No. of indicators	No. of pairs	Correlation value threshold ($\overline{\rho}$)			
			0.60	0.70	0.80	0.90
1 Connectivity	8	28	2	–	–	–
2 Human capital	6	15	8	7	4	3
3 Use of internet services	11	55	22	12	7	1
4 Integration of digital technology	7	21	7	5	1	–
5 Digital public services	5	10	2	1	–	–
Total	37	129	41	25	12	4

Table 4
Human capital: correlation matrix.

Indicators	2a1	2a2	2a3	2b1	2b2	2b3
2a1 At least basic digital skills	–					
2a2 Above basic digital skills	0.94	–				
2a3 At least basic software skills	1.00	0.92	–			
2b1 ICT specialists	0.77	0.75	0.75	–		
2b2 Female ICT specialists	0.56	0.62	0.53	0.87	–	
2b3 ICT graduates	0.11	0.20	0.09	0.37	0.45	–

Similar results are found when focusing on the category *Use of Internet Services* (Table 5), where indicators *People who never used internet* and *Internet Users* show a correlation equal to −0.97. Clearly, such finding is far from unexpected, as these indicators tackle the same aspect from two opposite perspectives. Nevertheless, their simultaneous inclusion in the DESI could produce a distortion in its final computation.

On a final note, less intense correlations are observed within the other categories, i.e., *Connectivity*, *Integration of digital technology* and *Digital Public Services*. Detailed results for these last three mentioned categories are available in the Appendix, Tables A1 – A3.

4.2. Step 2 – Principal Component Analysis

In this step, we applied PCA for each category, and determined the minimum number of PCs corresponding to an explained cumulative percentage variance at least equal to a given threshold ψ. Different scenarios were produced by varying $\psi \in \{0.70, 0.75, 0.8, 0.85, 0.90\}$. The obtained results are depicted in Fig. 1, which displays the number of PCs for each category guaranteeing the above discussed condition by the tested value of ψ.

For instance, we observe that 12 PCs are needed to explain at least 70 % of the cumulative variance ($\psi = 0.70$), four of which in the *Connectivity* category, two in the others. Note that, for each value of ψ, the most impacted categories, in terms of deleted indicators, are the ones that presented the higher redundancy from the correlation analysis (i.e., Human Capital and Use of Internet Services). Of course, the higher is the cumulative variance (ψ), the higher is the number of required PCs and of indicators to be considered in the reduced version of CI. Interestingly, we note that 25 PCs are obtained for $\psi = 0.95$, indicating that large pieces of information of the original DESI would be preserved even against a significant reduction in the number of indicators (25 out of 37). In the need of seeking a good trade–off between the number of PCs (hence, indicators) and the cumulative variance, we opted for a value of $\psi = 0.80$, corresponding to 15 PCs.

4.3. Step 3 – Indicators' selection and computation of the reduced index

Once determined the number of indicators to be included in the revised index (15), this step consists of (i) identifying the original indicators to consider and (ii) their weights. As discussed in Section 3.3, we select the indicators (without repetition for each category) having the highest correlation with the PCs obtained in Step 2. Then, the final version of the index was obtained by redistributing the weights of deleted indicators among the selected ones within the same sub-category. If a given sub-category is completely deleted, the associated weight is redistributed across the other sub-categories. The resulting revised version of the DESI is shown in Table 6.

It can be noticed that the reduction of indicators is not uniform across categories since it is influenced by the mutual correlation between their indicators (see Section 4.1). This is the reason why the number of indicators in the *Use of internet services* category is now reduced from 11 to 3. As we could expect, only the indicator *Internet users* has been included, while its "opposite" *People who never used internet* is deleted. A strong

Table 5
Use of internet services: correlation matrix.

Indicators	3a1	3a2	3b1	3b2	3b3	3b4	3b5	3b6	3c1	3c2	3c3
3a1 People who never used internet	–										
3a2 Internet users	−0.97	–									
3b2 News	−0.06	0.21	–								
3b2 Music, video and games	−0.40	0.54	0.53	–							
3b3 Video on demand	−0.70	0.75	0.09	0.69	–						
3b4 Video calls	0.36	−0.32	0.06	−0.06	−0.26	–					
3b5 Social networks	0.23	−0.12	0.31	0.10	−0.02	0.49	–				
3b6 Doing an online course	−0.69	0.73	0.22	0.67	0.84	−0.21	−0.04	–			
3c1 Banking	−0.80	0.84	0.42	0.55	0.60	−0.39	−0.13	0.64	–		
3c2 Shopping	−0.87	0.88	0.22	0.46	0.69	−0.54	−0.29	0.65	0.85	–	
3c3 Selling online	−0.66	0.67	0.13	0.50	0.75	−0.31	−0.13	0.59	0.63	0.79	–

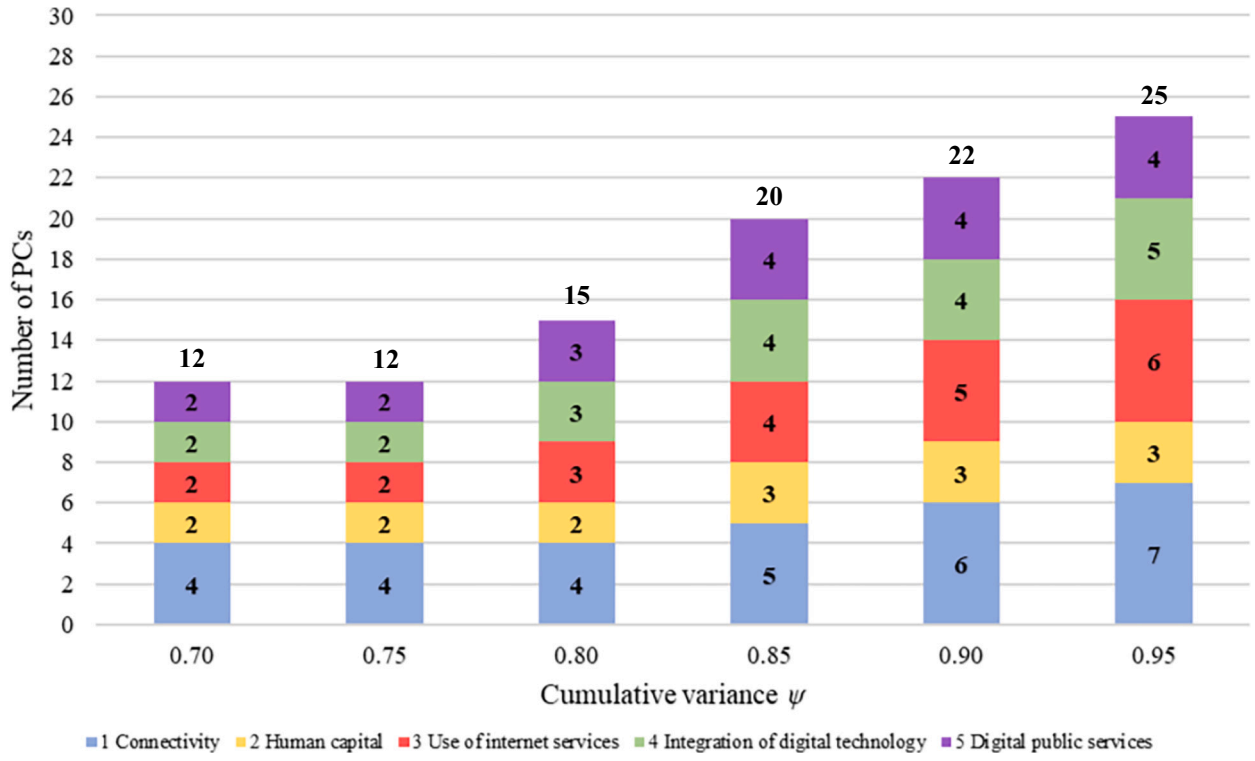


Fig. 1. Number of PCs by category and cumulative variance (ψ).

reduction is also observed for the *Human Capital* category, now comprising only two (out of six) indicators, namely *At least basic digital skills* and *ICT graduates* to measure non-specialist users' and more advanced users' digital skills, respectively. Among sub-categories, just 1d (*Broadband price index*) and 3c (*Transactions*) have not been selected in the reduced CIs.

4.4. Step 4 – Verification

We now need to understand to what extent the revised DESI is a good approximation of the original one. To this end, Fig. 2 compares the original and revised DESI calculated for all the 29 countries in our data set. As it can be observed, no significant differences are highlighted, indicating that the two versions yield similar values. This is also reflected by their high correlation, which equals 0.95, as shown in Table 7. In the first row of that table, correlations by category between the two indexes are reported. Here, the lowest value (0.75) is found for the *Use of Internet services* category, which likely depends on the relatively high number of deleted indicators. In general, the obtained high correlations seem to confirm that the reduced DESI can replicate the original one.

To further support this argument, it is worth noting that the produced rankings are very similar. Indeed, the Normalized Spearman's Footrule (NSF) value equals 0.88 (see Table 7, bottom row) and good results are obtained also when looking at the rankings' comparison by category.

Therefore, our empirical analysis let us conclude that: (i) the DESI suffers from some structural redundancies; (ii) our hypotheses and approach are validated, as the reduced index is reasonably consistent with the original DESI in terms of obtained results.

The latter finding is not of secondary importance. In fact, an agile version of the DESI may reduce data availability and reliability issues, enhancing its actual application at different geographical levels (especially within countries). An application to the Italian regions is discussed in the next section to give evidence of this fact.

5. An application to the Italian regions

This section shows an application of the reduced DESI version to the Italian regions. The data used and the obtained results are presented in Sections 5.1 and 5.2, respectively. Finally, we argue on the potential practical use and implications to policy-making in Section 5.3.

Table 6
The revised DESI framework.

Category [ID – Name (Weight)]	Sub-category [ID – Name (Weight)]	Indicators [ID – Name (Weight)]
1 Connectivity (25 %)	1a - Fixed Broadband take-up (0.312)	1a1 - Overall fixed BB take-up (1.000)
	1b - Fixed broadband coverage (0.312)	1b2 - Fixed VH Capacity Network coverage (1.000)
	1c - Mobile broadband (0.375)	1c2 - Mobile BB take-up (0.334)
		1c3 - 5G readiness (0.666)
2 Human capital (25 %)	2a - Internet User Skills (0.500)	2a1 - At least basic digital skills (1.000)
	2b - Advanced Skills and Development (0.500)	2b3 - ICT graduates (1.000)
3 Use of internet services (15 %)	3a - Internet use (0.334)	3a2 - Internet users (1.000)
	3b - Activities online (0.666)	3b1 - News (0.500)
		3b5 - Social networks (0.500)
4 Integration of digital technology (20 %)	4a - Business digitalization (0.600)	4a1 - Electronic information sharing (1.000)
	4b - e-Commerce (0.400)	4b1 - SMEs selling online (0.500)
		4b3 - Selling online cross-border (0.500)
5 Digital public services (15 %)	5a - e-Government (1.000)	5a1 - e-government users (0.333)
		5a3 - Online service completion (0.333)
		5a5 - Open data (0.333)

5.1. Test data

A necessary preliminary step for the analysis is to evaluate data availability at the regional level. Table 8 summarizes the results from this check, reporting, for each of the 37 original DESI indicators, the variable used for their calculation (according to the methodological note by the European Commission), whether they are selected in the proposed reduced DESI (✓) and available at the regional level (✓). For the sake of clarity, we underline that an indicator is available if its representative variable is retrievable from official data source for all the regions.

One interesting aspect emerging from the table is how the proposed methodology helps limiting the impact of missing data. Indeed, 19 out of 37 indicators (highlighted in cyan) of the original DESI would not be computable for our case study. This number, however, reduces to four (out of 15) considering the revised index (1c2, 2a1, 2b3, 5a3).

For these four not available indicators, appropriate proxy variables are introduced. Of course, the latter are chosen based on data availability considerations. Chosen proxy variables are shown in the Appendix, Table A4, which also informs on the official online sources used for downloading all the necessary data. All the used data can be made available to anyone interested upon request to the authors.

5.2. Results

European DESI 2020 scores (Fig. 3) show that Italy is the 26th country out of 29, occupying a low position in the ranking. This result testifies that, globally, Italy needs targeted actions to improve its digital conditions and reach the average European score.

However, understanding how regional digitalization levels impact

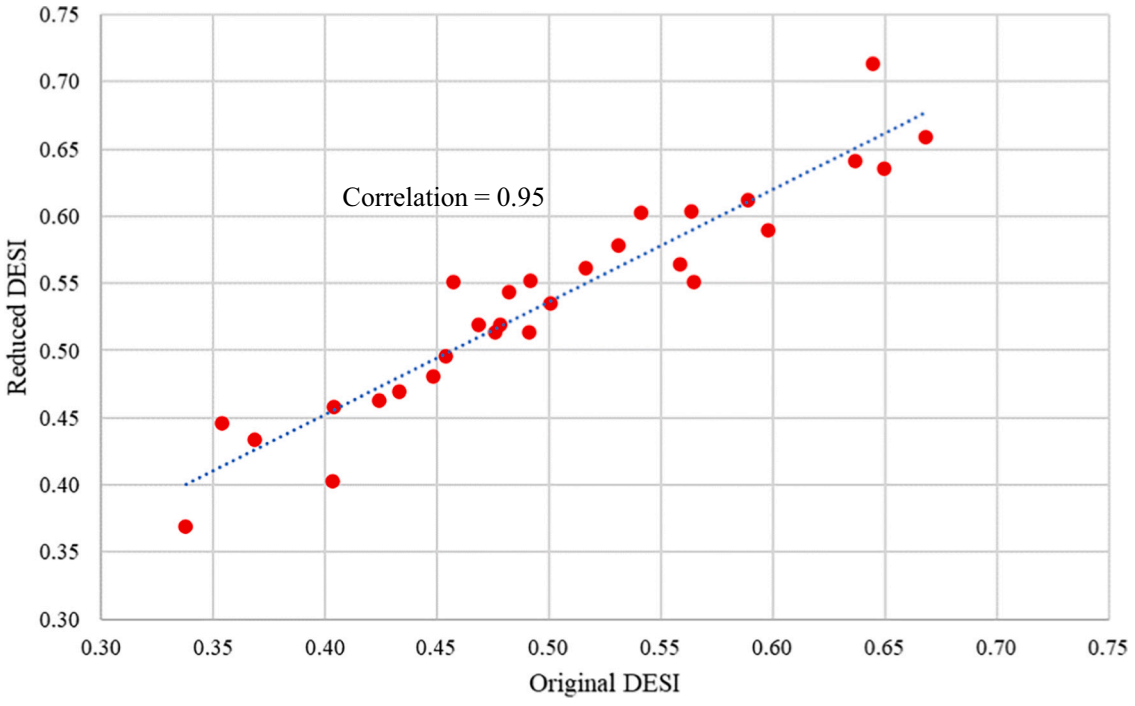


Fig. 2. Original vs. revised DESI: comparison of global scores.

Table 7
Original vs. revised DESI: correlations and Normalized Spearman's Footrule (NSF).

	DESI	Connectivity	Human Capital	Use of Internet services	Integration of digital technology	Digital public services
Correlation	0.95	0.93	0.87	0.75	0.80	0.88
NSF	0.88	0.83	0.77	0.74	0.71	0.79

Table 8
Data test used (Note: not available indicators highlighted in cyan).

Indicators	Selected in the reduced DESI	Availability
1a1 Overall fixed BB take-up	✓	✓
1a2 At least 100 Mbps fixed BB take-up		✓
1b1 Fast BB (NGA) coverage		✓
1b2 Fixed VH Capacity Network coverage	✓	✓
1c1 4G coverage		✓
1c2 Mobile BB take-up	✓	
1c3 5G readiness (double weighted)	✓	✓
1d1 Broadband price index		
2a1 At least basic digital skills	✓	
2a2 Above basic digital skills		
2a3 At least basic software skills		
2b1 ICT specialists		
2b2 Female ICT specialists		
2b3 ICT graduates	✓	
3a1 People who never used internet		
3a2 Internet users	✓	✓
3b1 News	✓	✓
3b2 Music, video and games		✓
3b3 Video on demand		
3b4 Video calls		
3b5 Social networks	✓	✓
3b6 Doing an online course		✓
3c1 Banking		✓
3c2 Shopping		
3c3 Selling online		✓
4a1 Electronic information sharing	✓	✓
4a2 Social media		
4a3 Big data (double weighted)		
4a4 Cloud (double weighted)		
4b1 SMEs selling online	✓	✓
4b2 e-commerce turnover		
4b3 Selling online cross-border	✓	✓
5a1 e-government users	✓	✓
5a2 Pre-filled forms		
5a3 Online service completion	✓	
5a4 Digital public services for business		
5a5 Open data	✓	✓

the overall national condition is essential to identify appropriate policies for digital conversion. Thus, the results obtained from applying the reduced DESI to the Italian regions can represent a meaningful support for policy-makers. In Fig. 4, reduced DESI scores, i.e., those calculated using the proposed reduced index, are shown for the Italian regions. The latter, indicated by their names and the official unique ID code provided by the Italian National Statistics Institute (ISTAT, see https://www.istat.it/), are ranked – from top to bottom – according to their overall DESI

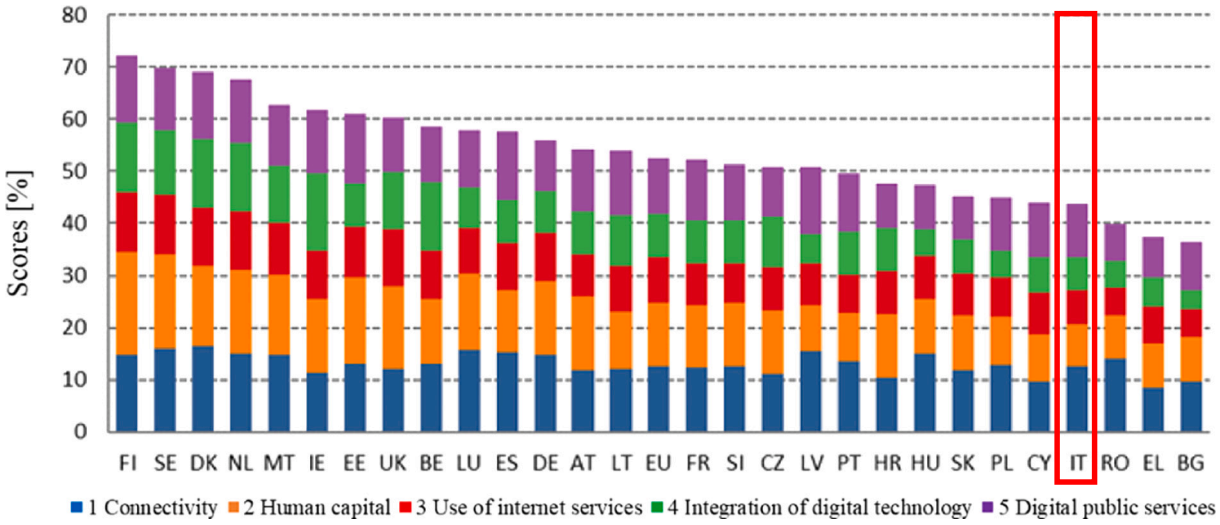


Fig. 3. Original DESI 2020 scores by country (Source: Desi 2020, European Commission).

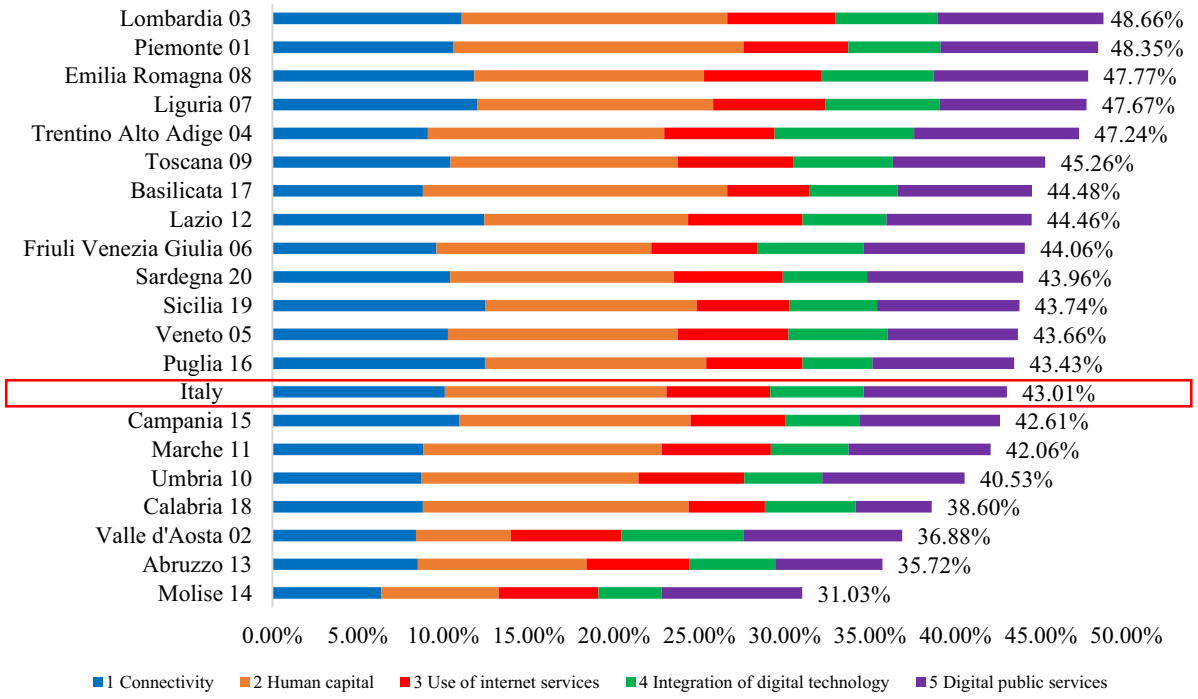


Fig. 4. Ranking of the Italian regions by the reduced DESI score.

score (reported at the end of each bar). Also, each bar displays the relative contribution of each category to the global score. Detailed results per region and indicator are instead reported in the Appendix, Table A5.

The above ranking highlights significative differences between regions, given the almost 20 % gap between the first and the last ranked ones. Notably, such differences are not negligible w.r.t. to the average Italian score, equal to 43.01 % (see the red box), especially for the last ranked regions. This fact proves the actual capability of the reduced index to capture local pattern of inequalities in DD measurement. Also, the produced results appear realistic and meaningful. Indeed, as Fig. 5 displays, regions showing the highest DESI scores are mainly concentrated in the northern area of the country, which is, in general terms, more industrialized and economically developed compared to southern regions. A higher degree of digital penetration and technological development in northern Italy regions is therefore a reasonable and expected result.

An interesting perspective in the analysis is given by comparing Italian regional and European rankings. Indeed, we note that Lombardia, the first ranked region, would occupy the 20th position, while Molise (the last one in the regional ranking) has a score of more than 5 % lower than the last ranked country. This observation reveals that, in general, all the Italian regions would need investments to improve their digitalization levels (if compared to the average European ones) and suggests a more in-depth analysis of the obtained results.

To this end, Fig. 6 shows the distributions of Italian regions, represented as points, by their DESI score per category. That picture also provides information on the average Italian and European scores (solid and dashed lines, respectively) and the ID of the best and the least performing regions per category. The figure can be analyzed from two

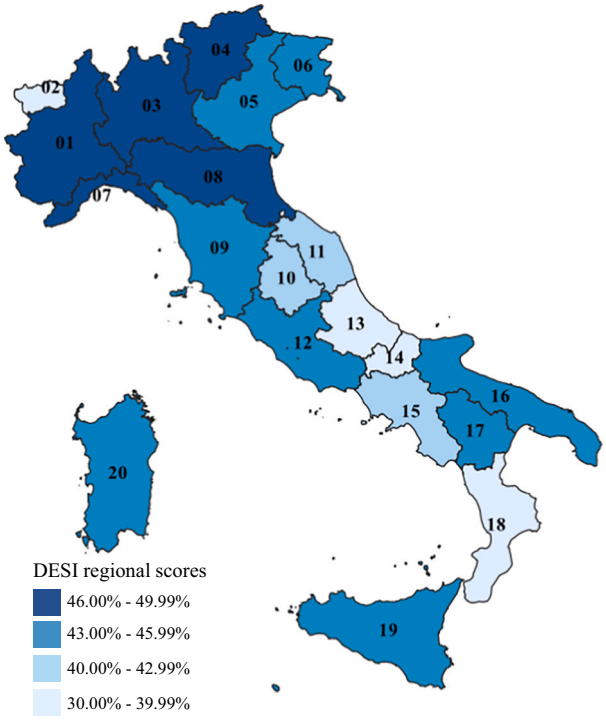


Fig. 5. Map of the Italian regions thematized according to reduced DESI's scores.

different perspectives. On the one hand, assuming the average European scores as benchmarks, their comparison with the Italian ones helps to understand whether actions should be implemented to improve the global Italian level in that category. On the other, the "compactness" of the distribution allows identifying those categories showing the highest DD between regions, thus requiring specific actions to reduce the gap.

It is possible to notice that the average Italian score is lower than the European one for four out of five categories (*Connectivity, Use of internet services, Integration of digital technology,* and *Digital public services*). Specifically, the first three categories also present relatively compact distributions, meaning that no significant gaps are observed among regions (except for a few outliers). This result indicates that the overall national level in these categories is lacking compared to the European one and that most Italian regions share such a condition. Under these circumstances, national policies to improve general digital conditions may be required. In particular, the category *Integration of digital technology* has the lowest average value. Such a finding suggests that actions involving the development of ICT digitalization in Italian companies are of utmost importance. Instead, the *Human capital* category shows an average value higher than the European one and a very dispersed distribution. In this case, the relevance of having an indicator applicable at the regional scale is clear. Indeed, when such circumstances occur, one can deduce that although the national condition is acceptable in average terms, the DD between regions is relevant and calls for specific local policies targeting the most lagging areas.

Based on these findings, a general framework to classify the DESI categories and derive potential policy actions is described next.

5.3. Policy implications

As highlighted above, the distribution of regional scores by category may provide useful indications to define appropriate policy actions. To support policy-makers in identifying them, one relatively simple option is to refer to the following two indicators:

(i) the *national average condition,* measuring the average performance of the country in a given category and defined as the regional average score μ by category;
(ii) the *inter-regional divide,* measuring the existing DD among regions and defined as the standard deviation σ of the regional scores' distribution by category.

Obviously, a higher value of μ indicates a better overall national level

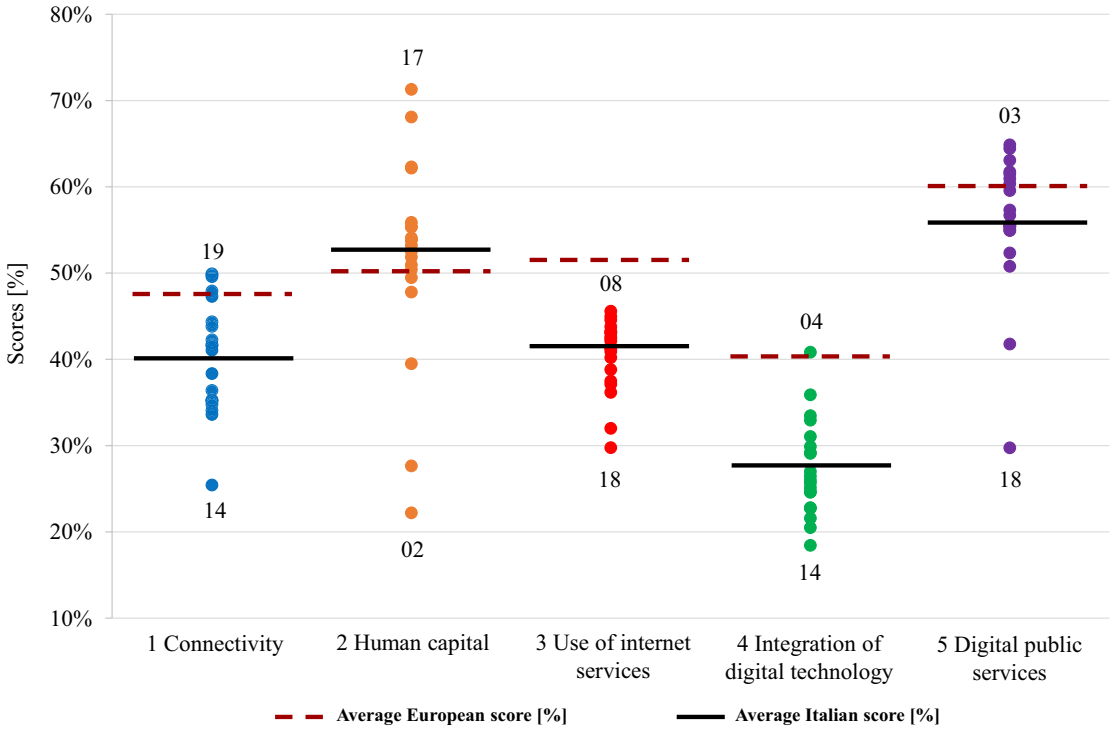


Fig. 6. Distributions of regional DESI scores by category.

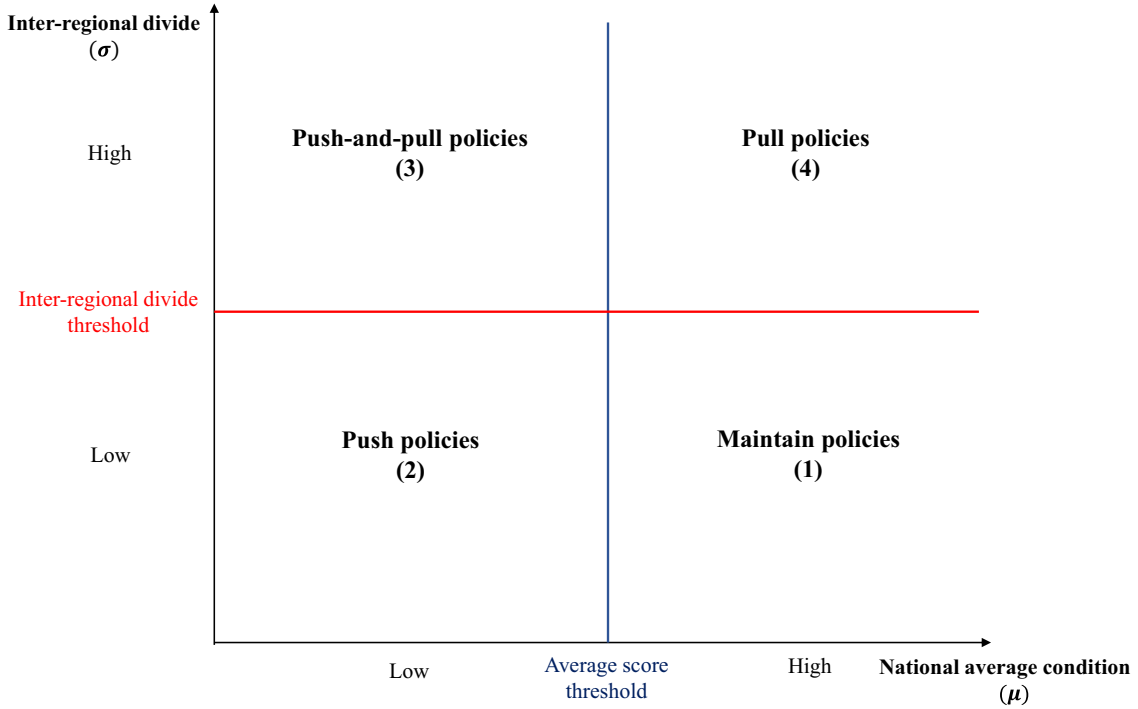


Fig. 7. DESI categories' classification and corresponding policies.

in a certain category, while a higher standard deviation σ indicates a more significant inter-regional divide.

By defining appropriate threshold/benchmark values for the above introduced indicators, it is possible to distinguish four areas of interests and define targeted policies, as shown in Fig. 7.

Specifically:

- *Group 1* involves categories characterized by a national condition that is over the threshold and low values of the inter-regional divide. This means that the national performance overcomes the reference threshold and that the divide among regions is somewhat limited. In practice, this is an ideal situation in which (almost) all the regions share satisfactory digitalization levels. Categories falling in this group would therefore require *maintenance policies*, i.e., actions

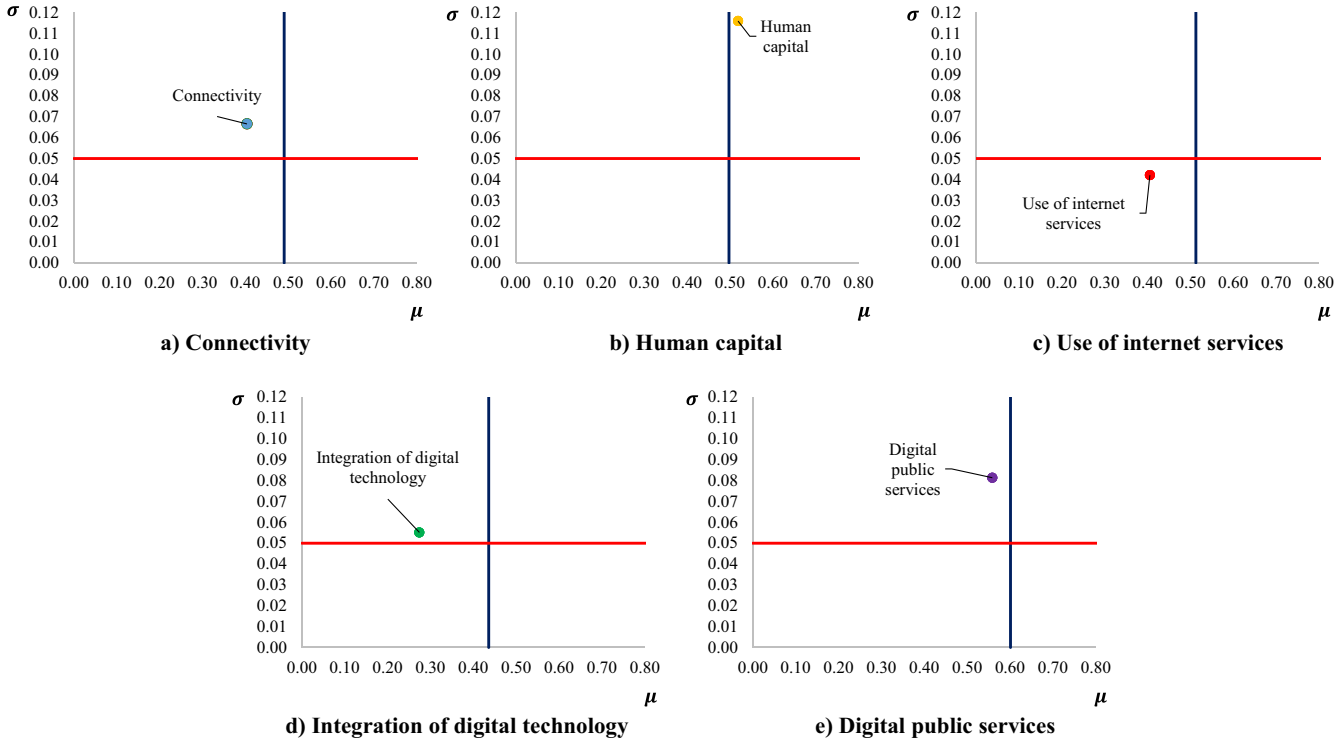


Fig. 8. Categories' classification for the Italian case study (μ is the average score, σ is the standard deviation).

aiming at consolidating and preserving the current condition (*maintain policies*).

- *Group 2* involves categories characterized by low values of both indexes. In this case, the national average condition in a given category does not reach the reference threshold. Also, the inter-regional gap is limited, thus meaning that regions share an unsatisfactory digitalization level. Categories belonging to this group need national interventions to *push* their development beyond acceptable minimum standards (*push policies*).
- *Group 3* involves categories characterized by an unsatisfactory national performance and a high inter-regional divide. In these circumstances, local and differentiated interventions are certainly required to *pull* less developed regions towards virtuous examples. This strategy is expected to contribute to a generalized improvement of the average national score, thus producing an indirect *push* effect. However, if not sufficient, further national policies should be required to accelerate the regions' growth and push all of them towards better performances (*push and pull policies*).
- Finally, categories in *group 4* exhibit high values of both indicators. The resulting interpretation is that the national level in a given category is satisfactory in average terms. Nevertheless, an alarming internal divide is observed, and attention should be paid to lagging regions. *Pull policies* emerge as appropriate in these cases, aiming to increase the digitalization level in the least performing regions and reducing internal disparities (*pull policies*).

Based on the above scheme, Fig. 8 depicts the classification obtained for the Italian case, by setting: (i) the average threshold value per category equal to the average European score in the category itself; (ii) the standard deviation threshold equal to 0.05. Note that these values are arbitrarily chosen for illustrative purposes. In practice, these parameters should be appropriately set by policy-makers depending on relevant benchmarks for the analysis.

According to the defined thresholds, the *Human capital* category would fall in group 4, i.e., *pull policies* would be required. In practice, priority should be given to interventions targeting the least performing regions, which can be easily identified in Fig. 6. In this specific case, for instance, the last positions held by Valle d'Aosta (ID 02) and Molise (ID 14) are explained by the low values of the *ICT graduates* indicator (equal to 0 and 0.20, respectively, see Table A5 in the Appendix). Therefore, raising awareness and understanding of engineering and science emerge as fundamental in such regions to promote and deliver Science, Technology, Engineering and Maths (STEM) education (proxy variable for *ICT graduates* indicator, see Table A4 in the Appendix), thus likely increasing the number of advanced ICT users and graduates. The *Use of internet services* category would instead belong to group 2, requiring *push policies* to improve the corresponding generalized low digitalization level. As indicators reveal (see Table A5 in the Appendix), the percentages of *internet users* are not negligible; yet, the number of them using the internet as an actual source of information is scarce (see the *News indicator*). This may raise, for instance, the need to improve users' sensitivity towards certain (informative) usages of web tools. Finally, *Connectivity*, *Integration of digital technology* and *Digital public services* categories would be included in group 3, calling for *push-and-pull policies* to improve the overall national level with a particular focus on lagging regions. Specifically, in terms of *connectivity*, infrastructural interventions seem necessary to achieve a spread diffusion of more performing connection technologies, e.g., *very high capacity networks* and *5G*. Note that *5G* does not cover at all various Italian regions (the corresponding indicator, *5G readiness,* equals 0 in Table A5). Besides, we observe that the poor national score in the *Integration of digital technology category* is mainly due to generalized low percentages of companies operating online in international markets (*selling online cross-border*) and, above all, adopting software solutions for *electronic information sharing* between their functional areas. Also, the high inter-regional divide is explainable by the presence of regions (especially the southern ones, like Molise – ID 14) with a relatively low number of businesses active in e-commerce (*SMEs selling online*). Possibly, incentive schemes at the national level should be designed to facilitate the adoption of digital solutions in companies. At the same, local support to help small businesses go digital is crucial to ease their survival in increasingly virtual markets.

Similar considerations can be drawn for the *Digital public service* category. In this case, the generalized national condition could be improved by enlarging and strengthening the spectrum of online services offered by public administrations, raising users' awareness towards their use (*e-government users*), and ensuring the actual availability of fully digitalized procedures in specific regions (*online services completion*).

6. Conclusions

The appropriate measurement of the DD is fundamental to describe inequalities in people's access and use of ICTs, and aid policy design driving the digital transformation of business and society. To this end, several CIs have been proposed over the years, characterized by a high number of indicators and complex structures. This often represents a limitation, especially for the reproducibility of the analysis at a lower territorial level.

In this context, the goal of the present paper was to define an easier-to-implement tool to evaluate DD among and, above all, within countries (i.e., at the regional level) to grasp sub-national divides and provide policy-makers with a useful tool to define appropriate policies to guide the digital transformation. With this aim in mind, we first analyzed the Digital Economy and Society Index (DESI), the most recent and adopted CI for DD measurement across European countries. In particular, we employed a PCA-based methodology to assess eventual redundancies in its structure. Using the 2020 data related to the 29 EU countries, we obtained a reduced version of the DESI, i.e., involving a significantly lower number of indicators (15 vs 37), capable of producing very similar results to the original one. The main implication of this agile and statistically robust version of the index is that it reduces the amount of needed data for its calculation and eases its application at different geographical levels. Through an application to the Italian case, we showed that the index can capture local patterns of digital inequalities between regions and that, depending on the scores' distributions, different policies can be defined to mitigate the internal divide and to improve or consolidate the overall country's digital level. In particular, a framework for classifying the DESI categories is proposed in an attempt to show how the revised index can support the identification of adequate interventions at both the national and regional levels and priority actions addressing areas suffering from digital exclusion.

The work done in this paper opens various research directions. From an application perspective, it would be interesting to apply the revised index to other case studies and compare the results obtained at the local level. This could inform a deeper discussion on a possible harmonization of policies to be undertaken at a country level and/or the setting of policy priorities in the EU digitalization agenda. From a methodological point of view, the analysis could be repeated by considering the data related to different years, in order to check if the detected redundancies still hold and to obtain a more robust version of the reduced index. Finally, it could be interesting to analyze the phenomenon of the DD from a dynamic perspective and to define a method to recognizing those indicators that, from year to year, are no longer significant for DD measurement. Therefore, further investigations should analyze the actual meaningfulness of single indicators to the DD, thus supporting the periodic update of the CI's structure by replacing obsolete indicators with new ones describing current and future dimensions of digitalization.

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CRediT authorship contribution statement

Giuseppe Bruno: Conceptualization, Methodology, Investigation, Validation. **Antonio Diglio:** Conceptualization, Methodology, Investigation, Validation. **Carmela Piccolo:** Conceptualization, Methodology, Investigation, Validation. **Eduardo Pipicelli:** Conceptualization, Methodology, Investigation, Validation.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

Appendix A

Table A1
Connectivity: correlation matrix.

	1a1	1a2	1b1	1b2	1c1	1c2	1c3
1a1 Overall fixed BB take-up	–						
1a2 At least 100 Mbps fixed BB take-up	0.20	–					
1b1 Fast BB (NGA) coverage	0.63	0.18	–				
1b2 Fixed VH Capacity Network coverage	0.12	0.67	0.14	–			
1c1 4G coverage	0.33	0.05	0.29	0.09	–		
1c2 Mobile BB take-up	−0.19	−0.07	−0.16	0.18	0.19	–	
1c3 5G readiness	−0.21	0.09	−0.07	−0.18	−0.04	−0.04	–

Table A2
Integration of digital technology: correlation matrix.

	4a1	4a2	4a3	4a4	4b1	4b2	4b3
4a1 Electronic information sharing	–						
4a2 Social media	0.39	–					
4a3 Big data	0.38	0.59	–				
4a4 Cloud	0.37	0.74	0.56	–			
4b1 SMEs selling online	0.34	0.64	0.47	0.73	–		
4b2 e-commerce turnover	0.22	0.47	0.31	0.64	0.82	–	
4b3 Selling online cross-border	0.37	0.53	0.37	0.46	0.78	0.70	–

Table A3
Digital public services: correlation matrix.

	5a1	5a2	5a3	5a4	5a5
5a1 e-government users	–				
5a2 Pre-filled forms	0.33	–			
5a3 Online service completion	0.35	0.74	–		
5a4 Digital public services for business	0.25	0.54	0.69	–	
5a5 Open data	0.19	0.07	0.04	0.13	–

Table A4
Proxy variables and online sources used for computing the revised DESI at the regional level (Note: not available indicators highlighted in cyan).

Categories	Indicators	Description	Sources	Proxy
1 - Connectivity	1a1 - Overall fixed BB take-up	% of households subscribing to fixed broadband	AGCOM https://www.agcom.it/	-
	1b2 - Fixed VH Capacity Network coverage	% of households covered by FTTH, FTTB.	AGCOM https://www.agcom.it/	-
	1c2 - Mobile BB take-up	% of mobile data subscriptions	AGCOM https://www.agcom.it/	% of households with mobile connection
	1c3 - 5G readiness	% of the amount of spectrum assigned and ready for 5G use.	AGCOM https://www.agcom.it/	-
2 - Human capital	2a1 - At least basic digital skills	% of individuals with 'basic' or 'above basic' digital skills in each of the following four dimensions: information, communication, problem solving and software for content creation	ISTAT https://www.istat.it/	% of enterprises whose employees have at least basic digital skills in each of the following four dimensions: information, communication, problem solving and software for content creation
	2b3 - ICT graduates	% of individuals with a degree in ICT	ISTAT https://www.istat.it/	% of individuals with a degree in STEM (science, technology, engineering and mathematics)
3 - Use of internet services	3a2 - Internet users	% of individuals who used the internet at least once a week (last three months)	ISTAT https://www.istat.it/	-
	3b1 - News	% of individuals who used the internet to read online news sites, newspapers or news magazines (last three months)	ISTAT https://www.istat.it/	-
	3b5 - Social networks	% of individuals who used the internet to participate in social networks (last three months)	ISTAT https://www.istat.it/	-
4 Integration of digital technology	4a1 - Electronic information sharing	% of businesses who have in use an ERP (enterprise resource planning) software package to share information between different functional areas	ISTAT https://www.istat.it/	-
	4b1 - SMEs selling online	% of SMEs selling online	ISTAT https://www.istat.it/	-
	4b3 - Selling online cross-border	% of SMEs that carried out electronic sales to other EU countries	ISTAT https://www.istat.it/	-
5 Digital public services	5a1 - E-government users	% of individuals who sent filled forms to public authorities over the internet in the previous 12 months	ISTAT https://www.istat.it/	-
	5a3 - Online service completion	Score that measures the share of administrative steps that can be done online for major life events (birth of a child, new residence, etc.)	ISTAT https://www.istat.it/	% of individuals who have electronic identity and electronic healthcare identity
	5a5 - Open data	Score that measures to what extent countries have an open data policy in place, the estimated political, social and economic impact of open data and the characteristics of the national data portal.	AGID https://www.agid.gov.it/	-

Table A5
Reduced DESI scores per region and indicator.

ID – Region	1 - Connectivity				2 - Human capital		3 - Use of internet services			4 - Integration of digital technology			5 - Digital public services		
	1a1-Overall fixed BB take-up	1b2 - Fixed VH Capacity Network coverage	1c2 - Mobile BB take-up	1c3 – 5G readiness	2a1 - At least basic digital skills	2b3 – ICT graduates	3a2 -Internet users	3b1 - News	3b5 - Social networks	4a1 -Electronic information sharing	4b1 - SMEs selling online	4b3 - Selling online cross-border	5a1 – E-government users	5a3 – Online service completion	5a5 - Open data
01 - Piemonte	0.84	0.33	0.36	0.05	0.37	0.99	0.55	0.39	0.29	0.23	0.38	0.28	0.27	0.90	0.68
02 - Valle d'Aosta	0.80	0.13	0.36	0.01	0.44	0.00	0.57	0.41	0.31	0.25	0.70	0.35	0.31	1.00	0.54
03 - Lombardia	0.91	0.32	0.35	0.06	0.37	0.88	0.62	0.38	0.26	0.26	0.44	0.28	0.26	0.85	0.83
04 - Trentino Alto Adige	0.81	0.20	0.40	0.00	0.34	0.77	0.64	0.42	0.23	0.28	0.98	0.24	0.31	0.93	0.69
05 - Veneto	0.86	0.24	0.40	0.07	0.34	0.74	0.60	0.42	0.28	0.25	0.47	0.25	0.28	0.66	0.58
06 - Friuli Venezia Giulia	0.80	0.27	0.41	0.00	0.36	0.65	0.60	0.42	0.22	0.26	0.46	0.33	0.29	0.89	0.72
07 - Liguria	0.90	0.47	0.40	0.01	0.35	0.76	0.57	0.41	0.34	0.21	0.66	0.38	0.27	0.70	0.75
08 - Emilia Romagna	0.83	0.37	0.42	0.19	0.34	0.74	0.62	0.43	0.31	0.26	0.52	0.34	0.26	0.94	0.61
09 - Toscana	0.86	0.31	0.34	0.04	0.32	0.75	0.59	0.37	0.38	0.20	0.63	0.23	0.24	0.93	0.62
10 - Umbria	0.68	0.25	0.40	0.03	0.36	0.66	0.56	0.37	0.32	0.20	0.37	0.18	0.23	0.83	0.60
11 - Marche	0.76	0.20	0.35	0.04	0.31	0.81	0.53	0.41	0.34	0.18	0.34	0.24	0.23	0.85	0.59
12 - Lazio	0.88	0.46	0.40	0.12	0.40	0.55	0.62	0.41	0.31	0.21	0.47	0.12	0.29	0.64	0.77
13 - Abruzzo	0.67	0.27	0.38	0.01	0.33	0.46	0.52	0.39	0.30	0.19	0.56	0.13	0.22	0.48	0.55
14 - Molise	0.57	0.11	0.34	0.00	0.36	0.20	0.47	0.38	0.32	0.19	0.27	0.07	0.23	0.93	0.50
15 - Campania	0.87	0.36	0.38	0.03	0.35	0.73	0.47	0.24	0.40	0.16	0.50	0.11	0.20	0.76	0.69
16 - Puglia	0.91	0.44	0.38	0.12	0.34	0.70	0.45	0.29	0.39	0.15	0.37	0.21	0.21	0.87	0.58
17 - Basilicata	0.73	0.23	0.41	0.00	0.35	1.07	0.41	0.27	0.28	0.25	0.44	0.10	0.22	0.73	0.62
18 - Calabria	0.76	0.16	0.40	0.06	0.38	0.87	0.43	0.19	0.28	0.15	0.74	0.12	0.16	0.18	0.55
19 - Sicilia	0.88	0.53	0.38	0.05	0.32	0.67	0.47	0.28	0.33	0.17	0.62	0.15	0.19	0.81	0.66
20 - Sardegna	0.88	0.25	0.45	0.02	0.36	0.69	0.51	0.44	0.33	0.22	0.50	0.07	0.28	0.91	0.64

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